

Notes: Campa and Kedia (JF, 2002)

Explaining the Diversification Discount

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1 Big picture

- **Research question.** Does the diversification discount show that diversification destroys value, or does it partly reflect endogenous self-selection by firms that choose to diversify or refocus?
- **Main idea.** Firms do not randomly become diversified. They diversify, stay focused, or refocus in response to firm and industry conditions. Those same conditions may also affect firm value. If the econometrician ignores this selection, the discount may be wrongly attributed to diversification itself.
- **Empirical object.** The paper studies excess value, defined as the log of actual firm value relative to imputed stand-alone value. The imputed value follows Berger and Ofek's method: each segment is valued using median single-segment firm multiples in its industry.
- **Baseline fact.** In standard OLS regressions, diversified firms trade at a significant discount. In the authors' sample, the multisegment discount is about 11% to 13%, similar to prior estimates.
- **Endogeneity concern.** Poorly performing firms, firms in declining industries, or firms searching for better organizational matches may be more likely to diversify. If these firms were already discounted before diversification, OLS overstates the causal value destruction from diversification.
- **Main finding.** Once endogeneity is modeled with firm fixed effects, instrumental variables, and Heckman-style self-selection corrections, the diversification discount falls sharply and sometimes becomes a premium.
- **Refocusing symmetry.** The paper also studies refocusing firms. For refocusing firms, selection works in the opposite direction: the characteristics that make firms refocus are positively correlated with firm value. Correcting for selection can make the estimated multisegment discount larger.
- **Economic takeaway.** The observed diversification discount is not necessarily causal evidence that diversification destroys value. Organizational form is a choice, and firm value must be analyzed jointly with the choice to diversify or refocus.

2 Incremental contribution of the paper

- **Challenges the causal interpretation of the diversification discount.** Earlier work documents that diversified firms have lower value than focused firms. Campa and Kedia argue that this fact alone does not prove that diversification causes value loss.
- **Explicitly models organizational form as endogenous.** The paper treats diversification status as an outcome chosen by firms, not as an exogenous treatment.
- **Uses multiple econometric approaches.** The analysis controls for endogeneity with three complementary methods: firm fixed effects, instrumental variables based on predicted diversification probability, and a Heckman self-selection correction.
- **Separates diversifying and refocusing firms.** The paper does not pool all conglomerates. It studies firms that diversify and firms that refocus separately, because the selection logic differs across the two decisions.

- **Reconciles discount and rational choice.** A firm can be discounted and still rationally choose diversification if its value would have been even lower had it remained focused. The paper makes this point empirically.
- **Documents pre-existing differences.** Single-segment years of firms that later diversify differ from always-focused firms. Industry exit rates, entry patterns, and firm characteristics are systematically related to later diversification and refocusing.
- **Shifts the interpretation of conglomerate evidence.** The paper suggests that the diversification discount is partly an equilibrium sorting outcome: firms with different opportunities and shocks choose different organizational forms.

3 Data and measurement

3.1 Sample selection

The sample is built from the Compustat Industry Segment database from 1978 to 1996. The paper follows the Berger–Ofek sample-selection rules and then adds additional cleaning filters.

- The final sample contains **8,815 firms** and **58,965 firm-years**.
- Firm-years are excluded if the firm reports a segment in the financial sector, has sales below \$20 million, has missing total capital, or has segment sales that differ from total sales by more than one percent.
- Firm-years are also excluded if the firm does not report four-digit SIC codes for all segments.
- Extreme excess values are removed when the log ratio of actual to imputed value exceeds 1.386 in absolute value.

3.2 Excess value

The central valuation measure is excess value:

$$\text{Excess Value}_{it} = \ln \left(\frac{\text{Actual Firm Value}_{it}}{\text{Imputed Stand-alone Value}_{it}} \right).$$

The imputed stand-alone value is the sum of the estimated values of the firm’s segments if each segment were operated as a focused firm. Segment values are computed using sales or asset multipliers of single-segment firms in the same industry.

- Negative excess value means the firm trades at a discount relative to comparable focused firms.
- Positive excess value means the firm trades at a premium.
- The paper reports results using both sales multipliers and asset multipliers.

3.3 Diversification profiles

The paper classifies firms by how their number of reported segments changes over time.

- Always single-segment firms form the largest group: 5,387 firms and 30,284 firm-years.
- 606 firms diversify during the sample period, generating 4,326 firm-years.
- 873 firms refocus during the sample period, generating 7,987 firm-years.
- 1,371 firms both focus and diversify at some point during the period.
- 578 multisegment firms do not change the number of reported segments.

3.4 Firm characteristics

The valuation regressions include standard firm-level variables:

- firm size, measured by log total assets;
- profitability, measured by EBIT to sales;
- investment, measured by capital expenditures to sales;
- leverage, measured by long-term debt to total assets;
- a nonlinear size term, the square of log total assets;
- lagged values of size, profitability, and investment.

3.5 Instruments for diversification and refocusing

The paper models the probability of multisegment operations using firm, industry, and macroeconomic variables.

- **Industry variables.** PNDIV is the fraction of firms in the industry that are conglomerates. PSDIV is the fraction of industry sales accounted for by conglomerates.
- **Merger-wave variables.** MNUM is the number of announced mergers and acquisitions in the year, and MVOL is their dollar value.
- **Macroeconomic variables.** GDP growth, lagged GDP growth, recession months, and lagged recession months proxy for business-cycle conditions.
- **Firm-market variables.** MAJOREX indicates listing on a major exchange, FOREIGN indicates foreign incorporation, and SNP indicates membership in the S&P industrial or transportation index.
- **Historical firm averages.** Historical average size, profitability, and investment summarize persistent firm characteristics.

4 Methodology and empirical specifications

4.1 Baseline valuation equation

The basic valuation equation is:

$$V_{it} = \delta_0 + \delta_1 X_{it} + \delta_2 D_{it} + e_{it},$$

where V_{it} is excess value, X_{it} is a vector of firm controls, and D_{it} equals one when the firm operates in multiple segments. In OLS, δ_2 is the estimated diversification discount.

- If $\delta_2 < 0$, multisegment firm-years trade at a discount.
- If $\delta_2 = 0$, multisegment status has no conditional valuation relation.
- If $\delta_2 > 0$, multisegment firm-years trade at a premium after controls.

4.2 Latent diversification decision

The paper models diversification status as a latent choice:

$$D_{it}^* = \beta' Z_{it} + u_{it},$$

with

$$D_{it} = 1 \quad \text{if } D_{it}^* > 0, \quad D_{it} = 0 \quad \text{otherwise.}$$

If unobserved determinants of D_{it} are correlated with unobserved determinants of V_{it} , OLS is biased.

4.3 Firm fixed effects

The first endogeneity control adds firm fixed effects and year effects. Firm fixed effects absorb persistent unobservable firm traits, such as managerial quality, organizational capabilities, or agency problems, that may affect both value and diversification choice.

Interpretation: if the discount falls after adding firm fixed effects, part of the OLS discount reflects persistent differences between firms rather than a causal effect of diversification.

4.4 Instrumental variables strategy

The paper first estimates a probit model for multisegment status using firm, industry, time, and macroeconomic predictors. The predicted probability of multisegment operations is then used as an instrument for the actual diversification dummy.

The logic is:

- industry and merger-wave variables help predict whether a firm diversifies or refocuses;
- excess value is already measured relative to industry-year medians, so industry-year shocks common to all firms should not mechanically determine relative value;
- the fitted probability isolates variation in organizational form that is related to observable determinants of the choice.

4.5 Heckman self-selection correction

The third approach uses a self-selection model. After estimating the probit, the paper computes a selection correction term and includes it in the excess-value regression:

$$V_{it} = \delta_0 + \delta_1 X_{it} + \delta_2 D_{it} + \delta_A A_{it} + \eta_{it}.$$

The coefficient on A_{it} captures the correlation between unobserved determinants of organizational choice and unobserved determinants of firm value.

- For diversifying firms, the expected selection coefficient is negative: firms more likely to diversify are also more likely to be discounted.
- For refocusing firms, the expected selection coefficient is positive: firms more likely to refocus are more likely to have high value opportunities.

4.6 Refocusing analysis

The paper repeats the strategy for refocusing firms. The refocusing sample includes single-segment firms and all refocusing firms. Here the question is whether multisegment operations among refocusing firms are more or less discounted once the choice to refocus is modeled.

Interpretation: if refocusing firms choose to refocus because staying diversified would be costly, then controlling for selection can increase the estimated discount associated with multisegment operations for these firms.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies whether the observed cross-sectional diversification discount survives controls for observable and persistent unobservable determinants of diversification status. It does not observe a randomized diversification treatment. The evidence is strongest as a correction to naive OLS and as a demonstration that selection is economically important.

5.2 Why excess value is useful

Excess value compares a firm to focused firms in the same industries and years. This is important because industries differ in valuation multiples and because aggregate market conditions vary over time. The measure is designed to remove industry and time shocks that affect all firms similarly.

5.3 Why OLS can be biased

OLS can be biased if firms that become diversified already differ from firms that stay focused. For example:

- a firm in an industry facing technological decline may diversify and be discounted even before diversification;
- a firm searching for a better organizational match may temporarily have lower value while it searches;
- a firm with high agency costs may both diversify and trade at a discount.

In all three cases, OLS can mistakenly attribute pre-existing discount to diversification.

5.4 Role of industry composition

The paper pays special attention to entry and exit. Exiting firms tend to have low excess value, while newly entering Compustat firms tend to have high excess value. Because industry medians change when firms enter and exit, the measured excess value of other firms can change mechanically. This matters if industry composition changes are correlated with diversification or refocusing decisions.

5.5 Interpreting the IV and selection results

For diversifying firms, the IV and self-selection models suggest that the OLS discount is too negative. In some specifications, the estimated effect of multisegment operations becomes positive. This supports the view that firms that choose to diversify are selected from firms that would otherwise have had low value.

For refocusing firms, the direction is different. Correcting for selection often makes the multisegment discount larger, especially with sales multipliers. This supports the view that firms that refocus are firms for which continued diversification would have been especially costly.

5.6 Limits of the evidence

The identification strategy depends on the validity of the instruments and the selection model. Industry and macro variables may still be correlated with relative value through channels not fully captured by excess-value construction. The Heckman model relies on distributional assumptions. Segment data are also imperfect and may reflect reporting changes, not only real changes in firm boundaries.

6 Tables and figures

6.1 Tables I and II: Baseline discount and diversification profiles

Read Table I:

- Table I replicates the standard diversification discount result using excess value as the dependent variable.
- In the Berger–Ofek-style specifications, the coefficient on the multisegment dummy is negative and highly significant.
- The extended full-sample model still implies an estimated discount of about 11% using both sales and asset multipliers.
- Larger and more profitable firms tend to have higher excess value, while the squared size term is negative in the extended model.

Read Table II:

- The sample contains 58,965 firm-years and 8,815 firms.
- Always-focused firms account for 30,284 firm-years.

- A large number of firms change diversification status: 606 diversify, 873 refocus, and 1,371 both focus and diversify.
- This distribution motivates treating diversification status as a dynamic choice rather than a fixed firm type.

Economic message: The standard discount exists in this sample, but many firms switch organizational form. The central question is therefore not whether diversified firms look discounted in OLS, but whether that discount is causal after accounting for why firms choose to diversify or refocus.

Table I
Estimation of the Diversification Discount

The dependent variable is excess value, defined as the log of the ratio of total market value to imputed value using median industry multipliers. The variable D is a dummy, which takes the value 1 when the firm operates in multisegments and 0 otherwise. The variable $EBIT/SALES$ is the ratio of $EBIT$ to sales, and $CAPX/SALES$ is the ratio of capital expenditures to sales. LEV is the ratio of long-term debt to total assets, and $ASS2$ is the square of the log of total assets. The second and fifth columns report the estimation of the BO(95) specification for the full sample, while columns 3 and 6 report the estimation of the BO(95) specification for the years 1986 to 1991. Columns 4 and 7 report the results of the extended model for the full sample (1978 to 1996). The t -statistics are reported in parentheses.

	Sales Multiplier			Asset Multiplier		
	BO 1978–1996	BO 1986–1991	Extended BO 1978–1996	BO 1978–1996	BO 1986–1991	Extended BO 1978–1996
Constant	−0.29 (39)	−0.32 (24)	−0.77 (36)	−0.07 (10)	−0.09 (7.76)	−0.45 (24)
D	−0.13 (26)	−0.12 (12)	−0.11 (20)	−0.12 (27)	−0.13 (14)	−0.11 (21)
Log of Total Assets	0.03 (22)	0.04 (17)	0.52 (43)	0.00 (0.35)	0.01 (3.67)	0.36 (33)
$CAPX/SALES$	0.33 (25)	0.29 (12)	0.17 (10)	0.05 (5)	0.03 (1.70)	−0.05 (3.77)
$EBIT/SALES$	1.05 (57)	0.95 (31)	0.86 (42)	0.82 (56)	0.73 (30)	0.76 (45)
Log of Total Assets (1 lag)			−0.16 (11)			−0.09 (−7.17)
$CAPX/SALES$ (1 lag)			0.05 (4.15)			0.01 (1.20)
$EBIT/SALES$ (1 lag)			0.00 (0.47)			0.00 (1.40)
Log of Total Assets (2 lag)			−0.19 (22)			−0.15 (20)
$CAPX/SALES$ (2 lag)			0.00 (0.86)			0.00 (1.35)
$EBIT/SALES$ (2 lag)			0.00 (0.71)			0.00 (0.64)
LEV			0.10 (8.26)			0.01 (0.78)
$ASS2$			−0.01 (17)			−0.01 (15)
Number of observations	54,451	16,429	45,291	52,126	15,524	43,194
Adjusted R^2	0.11	0.11	0.17	0.08	0.08	0.12

Table I: Estimation of the diversification discount.

Table II
Distribution of Firms by Diversification Profiles

This table reports the breakdown of firms in our sample by different diversification profiles. The sample consists of all firms reported in Compustat, from 1978 to 1996, subject to the sample selection criteria described in the text. All the data for the firm were used to classify firms into the categories (rather than just the years, which would be included after the sample selection criteria have been employed).

	Firm- years	Number of Firms
Firms that were always in single segments	30,284	5,387
Firms that diversified	4,326	606
Firms that diversified once from one segment to multiple segments	2,519	379
Firms that diversified once from multiple segments to multiple segments	791	105
Firms that diversified multiple times	1,016	122
Firms that refocused	7,987	873
Firms that refocused once from multiple segments to single segments	3,633	422
Firms that refocused once from multiple segments to multiple segments	1,588	183
Firms that refocused multiple times	2,766	268
Firms that both focused and diversified	13,133	1,371
Multisegment firms that did not change the number of segments	3,235	578
Total	58,965	8,815

Table II: Distribution of firms by diversification profiles.

6.2 Tables III and IVa: Firm characteristics and time patterns

Read Table III:

- Single-segment years of conglomerates differ from always-single-segment firms.
- Firms that diversify are larger, more levered, and often have different investment and profitability profiles.
- Multiple-segment years also differ across diversifying, refocusing, and non-changing conglomerates.
- These differences provide direct descriptive evidence that diversification status is not random.

Read Table IVa:

- Excess value varies substantially over time.

- Using sales multipliers, the median excess value is zero for several years in the late 1980s and early 1990s, but negative in many other years.
- The time pattern is important because entry, exit, and changes in the set of single-segment comparison firms affect imputed values.

Economic message: Firms with different diversification paths already look different before the main regressions. Time-varying comparison groups can also influence measured discounts, so industry composition matters for interpretation.

Table III
Summary Statistics by Diversification Profile

The table displays descriptive statistics for firms classified according to their diversification profile. Total assets are measured in millions of dollars. The value *CAPX/SALES* is the ratio of capital expenditures to total sales, *EBIT/SALES* is the ratio of EBIT to total sales, *DEBT/TA* is the ratio of long term debt to total assets, *INDGROW* is the growth in industry sales in the previous year, and *R&D/SALES* is the ratio of R&D expenditure to sales.

	Total Assets		<i>CAPX/SALES</i>		<i>EBIT/SALES</i>		<i>DEBT/TA</i>		<i>INDGROW</i>		<i>R&D/SALES</i>	
	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean	Median	Mean
Were always single segment firms	4.50	4.71	0.05	0.10	0.08	0.09	0.14	0.19	8.80	10.93	0.02	0.06
All firms that diversify	5.28	5.54	0.05	0.12	0.10	0.12	0.19	0.22	7.97	9.75	0.02	0.03
Single-segment years	5.01 ^a	5.27 ^a	0.05 ^a	0.12 ^a	0.10 ^a	0.13 ^a	0.17 ^a	0.21 ^a	8.80	10.67	0.02	0.04 ^a
Multiple-segment years	5.57	5.72	0.06	0.11	0.10	0.10	0.21	0.23	7.86	9.18	0.02	0.03
All firms that refocus	5.05	5.23	0.04	0.08	0.06	0.08	0.19	0.21	7.68	9.39	0.02	0.03
Single-segment years	4.82 ^a	5.02 ^a	0.04 ^a	0.09	0.07 ^a	0.08 ^a	0.18 ^a	0.21 ^a	8.05 ^a	9.62 ^a	0.02 ^a	0.04 ^a
Multiple-segment years	5.19 ^a	5.36 ^a	0.04 ^a	0.07 ^a	0.08 ^a	0.09 ^a	0.19 ^a	0.21 ^a	7.51 ^a	9.21	0.02 ^a	0.03 ^a
Multisegment firms with no change in diversification status	5.14 ^a	5.31 ^a	0.05 ^a	0.08 ^a	0.10	0.11 ^a	0.19 ^a	0.20 ^a	7.68	9.12	0.01 ^a	0.02 ^a
Firms that both focused and diversified	5.53	5.67	0.04	0.08	0.08	0.08	0.21	0.24	8.02	9.58	0.01	0.03
Single-segment years	4.86 ^a	5.12 ^a	0.04 ^a	0.09 ^a	0.08 ^a	0.09	0.22 ^a	0.25 ^a	8.06 ^a	9.84 ^a	0.01 ^a	0.03 ^a
Multiple-segment years	5.84 ^a	5.90 ^a	0.05 ^a	0.08 ^a	0.08 ^a	0.08 ^a	0.21	0.24	7.95	9.47	0.01 ^a	0.02 ^a

^a The difference with single-segment firms is significant at one percent.
^b The difference with single-segment firms is significant at five percent.
^c The difference with multiple-segment years of diversifying firms is significant at one percent.
^d The difference with multiple-segment years of diversifying firms is significant at five percent.

Table III: Summary statistics by diversification profile.

Table IVa
The Distribution of Excess Value over Time

This table displays summary statistics for the estimated excess value using sales and asset multipliers over time. The table includes all firm years, single segment as well as multiple segments.

	Sales Multiplier				Asset Multipliers			
	Median	Mean	Std. Dev.	Number	Median	Mean	Std. Dev.	Number
1978	-0.026	-0.044	0.493	2438	-0.017	-0.025	0.396	2281
1979	-0.040	-0.056	0.507	2393	-0.033	-0.053	0.430	2278
1980	-0.032	-0.057	0.542	2393	-0.028	-0.039	0.461	2257
1981	-0.057	-0.084	0.544	2527	-0.048	-0.078	0.456	2347
1982	-0.058	-0.074	0.549	2470	-0.040	-0.055	0.465	2327
1983	-0.052	-0.071	0.541	2643	-0.030	-0.045	0.453	2454
1984	-0.028	-0.045	0.538	2703	-0.022	-0.031	0.440	2536
1985	-0.030	-0.045	0.536	2690	-0.024	-0.025	0.446	2484
1986	-0.028	-0.041	0.539	2781	-0.031	-0.025	0.441	2578
1987	0.000	-0.013	0.553	2885	-0.017	-0.018	0.456	2729
1988	0.000	-0.022	0.543	2808	-0.005	-0.010	0.443	2693
1989	0.000	-0.009	0.553	2712	-0.002	-0.001	0.472	2632
1990	0.000	-0.028	0.578	2708	-0.006	-0.012	0.491	2626
1991	0.000	-0.033	0.576	2771	-0.011	-0.025	0.520	2697
1992	-0.017	-0.042	0.578	3049	-0.023	-0.026	0.524	2994
1993	-0.019	-0.039	0.572	3403	-0.024	-0.029	0.514	3332
1994	-0.020	-0.036	0.567	3682	-0.015	-0.029	0.495	3632
1995	-0.009	-0.027	0.583	3896	-0.011	-0.010	0.516	3895
1996	-0.005	-0.029	0.597	4182	-0.005	-0.013	0.522	4252

Table IVa: Distribution of excess value over time.

6.3 Tables IVb and IVc: Industry composition and exiting firms

Read Table IVb:

- Firms in their last year in the data tend to have negative excess value.
- Firms in their first year in the data tend to have positive excess value.
- Entry and exit can move industry medians, which affects the imputed values used to compute excess value.

Read Table IVc:

- Single-segment years of diversifying firms are in industries with high exit rates.
- Multiple-segment years of diversifying firms have lower exposure to exiting firms than their pre-diversification years.
- The pattern is consistent with firms moving away from troubled industries toward industries with better prospects.

Economic message: A firm can appear discounted after diversification partly because the set of comparison firms in its original industry changes. Industry exit is correlated with diversification decisions, so selection is not just a firm-level issue.

Table IVb

Effect of Changes in Industry Composition on Excess Value

The table reports results of excess value measured using sales multipliers. Column 2 shows mean excess values for single-segment firm-years. Median excess values are zero by construction. Column 3 shows the median excess values for the last year of firms in the data, column 4 shows the number of such firms, column 5 shows the median excess value for the first year of firms in the data, and column 6 shows the number of such firms.

Year	Mean	Median (for Last Year in Sample)	Number of Firms with Last Year	Median (for First Year in Sample)	Number of Firms with First Year
1978	0.0	0.019	78		
1979	-0.008	-0.026	73	0	106
1980	-0.005	0	66	0.098	107
1981	-0.015	-0.061	84	0	203
1982	-0.012	-0.063	81	0.112	116
1983	-0.000	-0.165	134	0.135	234
1984	0.011	-0.037	128	0.089	227
1985	0.011	-0.024	147	0.134	201
1986	-0.004	-0.041	152	0.105	274
1987	0.006	-0.14	211	0.072	269
1988	-0.006	-0.126	179	0.139	206
1989	0.011	-0.011	117	0.129	175
1990	-0.009	-0.197	119	0.078	158
1991	-0.011	-0.286	90	0.207	225
1992	-0.019	-0.15	117	0.259	280
1993	-0.009	-0.112	175	0.159	394
1994	-0.009	-0.063	253	0.109	397
1995	-0.013	-0.125	271	0.144	433
1996	-0.013			0.158	500

Table IVb: Effect of changes in industry composition
on excess value.

Table IVc

Summary Statistics for the Distribution of Exiting Firms

This table displays the distribution of firms that exit the sample (moved to the research files in the COMPUSTAT database). The percentage share of these firms in sales is the percentage of overall industry sales in a given year (defined at the two-digit SIC level) accounted for by all exiting firms in the sample. For a conglomerate, we take the sales weighted average of each segment. Similarly, the percentage of exiting firms in the industry is the fraction of all firms in the industry (defined at the two-digit SIC) in a given year accounted for by firms in the research file.

	Percent Share of Exiting Firms in Industry Sales				Percentage of Exiting Firms in Industry			
	Median	Mean	Std. Dev.	Number	Median	Mean	Std. Dev.	Number
single-segment firms	6.44	11.22	13.2	21,794	14.28	19.87	18.7	21,794
single-segment years of conglomerates	9.74	14.00	13.9	6,615	21.94	24.94	18.8	6,615
multiple-segment years of conglomerates	13.09	15.75	13.2	13,831	27.28	28.63	19.0	13,831
diversifying firms	9.65	13.31	12.7	3,287	21.71	24.07	17.9	3,287
single-segment years of diversifying firm	15.14	17.60	13.4	1,424	28.47	31.34	16.5	1,424
multiple-segment years of diversifying firm	6.13	10.04	11.1	1,863	14.53	18.59	16.9	1,863
focusing firms	11.56	14.91	13.2	4,934	26.35	27.74	19.2	4,934
single-segment years of focusing firms	5.82	9.89	11.2	2,024	14.52	19.12	17.1	2,024
multiple-segment years of focusing firms	17.02	18.40	13.3	291	37.00	33.7	18.2	291
firms that both diversify and focus	13.19	16.12	13.8	10,444	27.84	28.79	19.1	10,444
single-segment years	10.80	15.00	15.0	3,167	22.78	25.84	19.6	3,167
multiple-segment years	14.13	16.60	13.2	7,277	30.36	30.08	18.8	7,277
conglomerates that did not change diversification status	9.87	13.95	12.9	1,781	21.82	24.88	18.9	1,781

Table IVc: Summary statistics for the distribution of
exiting firms.

6.4 Tables IVd and V: Exiting firms and diversifying-firm regressions

Read Table IVd:

- Single-segment years of diversifying firms do not look deeply discounted relative to the industry median.
- They are valued above single-segment firms that exit the same industry.
- They are valued below single-segment firms that remain in the industry.
- This supports the view that diversifying firms come from weaker parts of their industries but are not the worst firms.

Read Table V:

- In the diversifying-firm sample, OLS estimates a significant discount around 9% to 13%.
- Firm fixed effects reduce the discount to about 6% using sales multipliers and 4% using asset multipliers.
- IV estimates turn the multisegment coefficient positive: 0.30 using sales multipliers and 0.19 using asset multipliers.
- The self-selection model also reduces or eliminates the discount; the selection coefficient is negative, consistent with firms more likely to diversify also being more discounted.

Economic message: The diversifying-firm results are the heart of the paper. Once the choice to diversify is modeled, the apparent discount is much weaker and may become a premium.

Table IVd
Exiting Firms and Excess Value

This table documents summary statistics for excess value of single-segment years of diversifying firms that did not exit, single-segment firms in the same industry that did not exit and did not diversify, and single-segment years of firms in the same industries that exited. It also displays the above for firms that refocus.

	Sales Multipliers				Asset Multipliers			
	Median	Mean	Std. Dev.	Number	Median	Mean	Std. Dev.	Number
Single-segment firms that do not exit ^a	0.02	0.04	0.57	11,771	0.012	0.05	0.5	12,279
Single-segment years of diversifying firms	0.0	-0.004	0.53	1,327	0	0.02	0.45	1,390
Single-segment firms that exit ^b	-0.04	-0.08	0.56	5,511	-0.03	-0.05	0.48	5,736
Single-segment firms that do not exit ^c	0.02	0.04	0.57	12,442	0.01	0.05	0.5	12,946
Single-segment years of focusing firms	0.0	-0.01	0.54	1,919	0.0	0.0001	0.47	1,992
Single-segment firms that exit ^d	-0.07	-0.10	0.54	5,610	-0.05	-0.07	0.47	5,825

^aThis group consists of single-segment firms that did not exit, that is, were not included in the research tapes in Compustat. In addition, these firms were in the same four-digit SIC as the single-segment years of diversifying firms.

^bThis group consists of single-segment years of firms that exited the sample and were in the same four-digit SIC as the single-segment years of diversifying firms.

^cThis group consists of single-segment firms that did not exit and were in the same four-digit SIC as the single-segment years of focusing firms.

^dThis group consists of single-segment years of firms that exited the sample and were in the same four-digit SIC as the single-segment years of focusing firms.

Table IVd: Exiting firms and excess value.

Table V
Effect of Multiple Segment Operations on Firm Value of Diversifying Firms

The sample consists of single-segment firms and all diversifying firms. The dependent variable is excess value, which is the ratio of firm value to its imputed value. The variable *D* is a dummy that takes the value 1 when the firm operates in multisegments and 0 otherwise. *EBIT/SALES* is the ratio of *EBIT* to sales, and *CAPX/SALES* is the ratio of capital expenditures to sales. The variable *SNP* is a dummy that takes the value of 1 when the firm is part of the S&P index and 0 otherwise. *LEV* is the ratio of long-term debt to total assets, and *ASS2* is the square of the log of total assets. Columns 2 to 6 (columns 7 to 11) report results when sales (asset) multipliers are used to calculate imputed value. Year dummies were included in the instrumental variable (IV) model as well as the self-selection model and are not reported. The fixed-effects model has two-way fixed effects. The standard errors are corrected for autocorrelation, and the corresponding *T*-statistics are given in parentheses below.

	Sales Multipliers					Asset Multipliers				
	OLS (BO)	OLS	Fixed Effects	IV	Self-selection	OLS (BO)	OLS	Fixed Effects	IV	Self-selection
Constant	-0.36 (33)	-0.75 (26)	-0.09 (1.67)	-0.72 (30)	-0.68 (30)	-0.14 (14)	-0.43 (17)	0.68 (14)	-0.34 (15)	-0.32 (15)
<i>D</i>	-0.13 (10)	-0.11 (9.13)	-0.06 (2.88)	0.30 (5.03)	0.18 (4.03)	-0.11 (9.09)	-0.09 (7.94)	-0.04 (2.08)	0.19 (3.29)	0.01 (0.24)
Log of total assets	0.04 (19)	0.61 (36)	0.33 (16)	0.55 (40)	0.54 (39)	0.01 (4.75)	0.42 (28)	0.00 (0.24)	0.24 (19)	0.24 (18)
<i>EBIT/SALES</i>	1.15 (42)	0.69 (19)	0.39 (13)	0.44 (13)	0.44 (13)	0.98 (44)	0.77 (26)	0.71 (27)	0.63 (16)	0.62 (15)
<i>CAPX/SALES</i>	0.33 (15)	0.06 (1.96)	0.19 (7.25)	0.16 (5.65)	0.16 (5.68)	0.04 (2.22)	-0.06 (2.38)	-0.03 (1.44)	-0.02 (0.98)	-0.02 (0.90)
Log of TA (1 lag)	-0.25 (12)	-0.28 (19)	-0.28 (13)	-0.25 (13)	-0.25 (13)	-0.15 (8.10)	-0.12 (8.75)	-0.04 (2.32)	-0.04 (2.50)	-0.04 (2.50)
<i>EBIT/SALES</i> (1 lag)	0.20 (5.57)	0.16 (5.99)	0.20 (5.49)	0.20 (5.42)	0.20 (5.42)	0.07 (2.18)	0.14 (6.11)	0.10 (2.31)	0.10 (2.28)	0.10 (2.28)
<i>CAPX/SALES</i> (1 lag)	0.08 (2.45)	0.05 (2.04)	0.09 (3.51)	0.09 (3.40)	0.09 (3.40)	-0.05 (1.91)	-0.04 (1.89)	-0.01 (0.67)	-0.02 (0.86)	-0.02 (0.86)
Log of TA (2 lag)	-0.20 (16)	-0.14 (14)	-0.14 (13)	-0.14 (13)	-0.14 (13)	-0.17 (16)	-0.11 (13)	-0.12 (12)	-0.12 (12)	-0.12 (12)
<i>EBIT/SALES</i> (2 lag)	0.09 (3.58)	0.09 (4.84)	0.08 (3.14)	0.08 (3.19)	0.08 (3.19)	-0.02 (1.06)	0.05 (3.02)	-0.003 (0.11)	-0.002 (0.09)	-0.002 (0.09)
<i>CAPX/SALES</i> (2 lag)	0.08 (3.65)	0.06 (3.08)	0.07 (3.94)	0.07 (3.80)	0.07 (3.80)	0.05 (2.78)	0.02 (1.08)	0.04 (2.79)	0.04 (2.65)	0.04 (2.65)
<i>SNP</i>			0.25 (27)	0.25 (28)	0.25 (28)			0.29 (33)	0.28 (33)	0.28 (33)
<i>LEV</i>	-0.03 (1.48)	0.22 (10)	0.09 (6.81)	0.11 (8.11)	0.11 (8.11)	-0.06 (3.94)	0.12 (5.72)	0.03 (2.21)	0.03 (3.61)	0.04 (3.61)
<i>ASS2</i>	-0.01 (11)	0.01 (6.12)	-0.01 (19)	-0.01 (19)	-0.01 (19)	-0.01 (10)	0.01 (6.38)	-0.01 (16)	-0.01 (16)	-0.01 (16)
Lambda				-0.14 (6.07)	-0.14 (6.07)				-0.04 (1.83)	-0.04 (1.83)
Hausman test (<i>P</i> -value)				43 (0.0)	43 (0.0)				20 (0.0)	20 (0.0)
<i>T</i> -statistics: first stage				22.0	22.0				21.0	21.0
Partial <i>R</i> ² : first stage				0.018	0.018				0.018	0.018
No. of observations	24,964	24,964	24,964	24,964	24,964	25,480	25,480	25,480	25,480	25,480
Adjusted <i>R</i> ²	0.13	0.20	0.65	0.14	0.14	0.09	0.14	0.59	0.08	0.08
<i>F</i> -statistic	932	518	13.05	138	135	607	350	10.20	73	72

Table V: Effect of multisegment operations on firm value of diversifying firms.

6.5 Tables VI and VII: Probit for diversification and refocusing-value regressions

Read Table VI:

- The probit predicts whether a firm operates in multiple segments in the diversifying-firm sample.
- PNDIV is positive and strongly significant: firms in industries with more conglomerates are more likely to diversify.
- Merger activity, measured by MNUM, increases the probability of multisegment operations.
- Major-exchange listing increases the probability of diversification, while S&P membership is negatively related to diversification.

Read Table VII:

- For refocusing firms, the baseline multisegment discount is large and significant.
- Unlike the diversifying-firm case, the fixed-effects estimates do not make the discount disappear.
- IV and selection corrections with sales multipliers make the estimated discount larger, about 21%.

- The selection coefficient is positive with sales multipliers, consistent with refocusing firms having unobserved traits positively related to value.

Economic message: Diversification and refocusing are not symmetric in OLS. Firms that diversify are negatively selected relative to value; firms that refocus appear positively selected. Correcting selection changes the estimated effect in opposite directions.

Table VI
Probit Estimates for Diversifying Firms

The dependent variable takes the value 1 when the firm operates in multiple segments and 0 otherwise. The reported results are for a sample of single-segment firms and all diversifying firms included in the excess value regressions using sales multipliers. The value *EBIT/SALES* is the ratio of *EBIT* to sales, and *CAPX/SALES* is the ratio of capital expenditures to sales. The variables *A_TA*, *A_EBIT*, and *A_CAPX* are the historical average values of the log of *TA*, *EBIT/SALES*, and *CAPX/SALES*. The variable *PNDIV* is the fraction of all firms in the industry that are conglomerates, and *PSDIV* is the fraction of industry sales accounted for by conglomerates. The variable *MNUM* is the number of announced mergers/acquisitions in the year, while *MVOL* is the U.S. dollar value of these. *GDP (GDPI)* is the growth rate in real *GDP* (and its lagged value), while the variable *CONTRACTION (CONTRACTION1)* is the number of months in the year the economy was in a recession (and its lagged value). The variable *MAJOREX* is a dummy that takes the value 1 if the firm is listed on Nasdaq, NYSE, or AMEX, and 0 otherwise. The variable *FOREIGN* is a dummy that takes the value 1 if the firm is incorporated outside the United States and 0 otherwise. The variable *SNP* is a dummy that takes the value 1 when the firm is part of the S&P index and 0 otherwise.

	Coefficient	T-statistic	Marginal Effect
Constant	-3.47	-31.03	-0.21
Log of total assets	0.22	3.98	0.03
<i>EBIT/SALES</i>	-0.07	-0.50	-0.01
<i>CAPX/SALES</i>	0.03	0.29	0.00
Log of <i>TA</i> (1 lag)	-0.10	-1.19	-0.01
<i>EBIT/SALES</i> (1 lag)	-0.07	-0.48	-0.01
<i>CAPX/SALES</i> (1 lag)	-0.11	-0.80	-0.02
Log of <i>TA</i> (2 lag)	0.23	4.18	0.02
<i>EBIT/SALES</i> (2 lag)	-0.10	-1.02	-0.01
<i>CAPX/SALES</i> (2 lag)	-0.09	-0.87	-0.02
<i>SNP</i>	-0.19	-4.08	-0.03
<i>PNDIV</i>	0.02	15.85	0.003
<i>PSDIV</i>	0.001	1.45	0.0002
<i>MVOL</i>	-0.0001	-0.36	0.00
<i>MNUM</i>	0.11	3.69	0.01
<i>GDP</i>	-0.01	-1.26	-0.002
<i>GDPI</i>	0.004	0.36	0.001
<i>CONTRACTION</i>	-0.08	-0.80	-0.02
<i>CONTRACTION1</i>	-0.02	-0.24	-0.01
<i>A_TA</i>	-0.20	-6.15	-0.02
<i>A_EBIT</i>	0.10	1.35	0.01
<i>A_CAPX</i>	0.04	0.39	0.01
<i>MAJOREX</i>	0.16	6.02	0.02
<i>FOREIGN</i>	0.06	1.22	0.02
No. of observations			24,964
Maximized likelihood			-6.478
Likelihood index			0.08
% multisegment observations			8.13

Table VI: Probit estimates for diversifying firms.

Table VII
Effect of Multiple-Segment Operations on Firm Value of Refocusing Firms

The sample consists of single-segment firms and all refocusing firms. The dependent variable is excess value, which is the ratio of firm value to its imputed value. The variable *D* is a dummy that takes the value 1 when the firm operates in multisegments and 0 otherwise. The value *EBIT/SALES* is the ratio of *EBIT* to sales, and *CAPX/SALES* is the ratio of capital expenditures to sales. The variable *SNP* is a dummy that takes the value 1 when the firm is part of the S&P index and 0 otherwise. The variable *LEV* is the ratio of long-term debt to total assets, and *ASS2* is the square of the log of total assets. Columns 2 to 6 (7 to 11) report results when sales (asset) multipliers are used to calculate imputed value. Year dummies have been included for the estimation of the instrumental variable (IV) model as well as that of the self-selection model. Standard errors have been corrected for autocorrelation, and *T*-statistics are reported in parentheses below.

	Sales Multipliers					Asset Multipliers				
	OLS (BO)	OLS	Fixed Effects	IV	Self- selection	OLS (BO)	OLS	Fixed Effects	IV	Self- selection
Constant	-0.38 (36)	-0.76 (28)	-0.14 (2.82)	-0.68 (32)	-0.68 (32)	-0.17 (18)	-0.43 (17)	0.62 (13)	-0.31 (15)	-0.31 (15)
<i>D</i>	-0.17 (17)	-0.13 (14)	-0.14 (9.20)	-0.21 (9.60)	-0.21 (10)	-0.13 (14)	-0.11 (12)	-0.11 (7.05)	-0.10 (4.51)	-0.10 (4.48)
Log of total assets	0.05 (22)	0.59 (36)	0.35 (17)	0.53 (41)	0.53 (41)	0.01 (7.23)	0.39 (28)	0.01 (0.69)	0.22 (18)	0.22 (18)
<i>EBIT/SALES</i>	1.14 (43)	0.66 (19)	0.37 (13)	0.40 (12)	0.40 (12)	1.00 (46)	0.76 (27)	0.67 (28)	0.59 (15)	0.59 (16)
<i>CAPX/SALES</i>	0.32 (15)	0.10 (3.43)	0.18 (7.46)	0.17 (6.64)	0.17 (6.64)	0.04 (2.44)	-0.03 (1.30)	-0.03 (1.24)	-0.004 (0.21)	-0.004 (0.20)
Log of <i>TA</i> (1 lag)	-0.24 (12)	-0.27 (19)	-0.24 (14)	-0.24 (14)	-0.24 (14)	-0.14 (8.09)	-0.12 (9.43)	-0.04 (2.79)	-0.04 (2.78)	-0.04 (2.78)
<i>EBIT/SALES</i> (1 lag)	0.26 (7.15)	0.21 (7.60)	0.23 (6.25)	0.23 (6.25)	0.23 (6.25)	0.13 (4.38)	0.20 (8.74)	0.16 (3.81)	0.16 (3.84)	0.16 (3.84)
<i>CAPX/SALES</i> (1 lag)	0.03 (0.83)	0.04 (1.91)	0.05 (2.03)	0.05 (2.04)	0.05 (2.04)	-0.07 (2.79)	-0.04 (2.19)	-0.03 (1.59)	-0.03 (1.60)	-0.03 (1.60)
Log of <i>TA</i> (2 lag)	-0.18 (15)	-0.12 (13)	-0.11 (11)	-0.12 (11)	-0.12 (11)	-0.15 (15)	-0.10 (12)	-0.10 (11)	-0.10 (11)	-0.10 (11)
<i>EBIT/SALES</i> (2 lag)	0.12 (4.60)	0.12 (5.86)	0.09 (3.48)	0.10 (3.56)	0.10 (3.56)	-0.02 (1.10)	0.05 (2.76)	-0.01 (0.28)	-0.01 (0.24)	-0.01 (0.24)
<i>CAPX/SALES</i> (2 lag)	0.09 (3.72)	0.06 (3.16)	0.06 (3.23)	0.06 (3.22)	0.06 (3.22)	0.06 (2.93)	0.02 (1.49)	0.04 (2.46)	0.04 (2.46)	0.04 (2.46)
<i>SNP</i>			0.25 (29)	0.25 (29)	0.25 (29)			0.28 (35)	0.28 (35)	0.28 (35)
<i>LEV</i>		0.02 (1.43)	0.25 (12)	0.17 (13)	0.17 (12)		-0.04 (2.58)	0.15 (7.59)	0.09 (7.23)	0.08 (7.05)
<i>ASS2</i>		-0.01 (12)	0.01 (3.48)	-0.01 (21)	-0.01 (21)		-0.01 (-9.85)	0.01 (5.76)	-0.01 (15)	-0.01 (15)
Lambda					0.03 (2.44)					-0.01 (0.48)
Hausman test (<i>P</i> -value)				3.20 (0.07)					0.08 (0.45)	
<i>T</i> -statistics: first stage				52					47	
Partial <i>R</i> ² : first stage				0.079					0.066	
No. of observations	27,995	27,995	27,995	27,995	27,995	28,161	28,161	28,161	28,161	28,161
Adjusted <i>R</i> ²	0.13	0.19	0.64	0.13	0.14	0.09	0.14	0.58	0.08	0.08
<i>F</i> -statistic	1064	557	13.00	142	142	732	383	10.30	77	77

Table VII: Effect of multisegment operations on firm value of refocusing firms.

6.6 Table VIII: Probit estimates for refocusing firms

Read:

- The refocusing probit fits better than the diversification probit, with a likelihood index of 0.21.
- Firm characteristics matter more for refocusing: historical size, historical profitability, and historical investment help explain multisegment status.
- *PNDIV* and *PSDIV* are positive and significant, meaning that industries attractive to conglomerates are associated with lower refocusing probability.
- *MNUM* is negative, suggesting that favorable merger markets encourage refocusing as well as diversification.

- Macroeconomic conditions significantly predict refocusing decisions.

Economic message: The decision to refocus is strongly related to observable firm, industry, merger-market, and macroeconomic conditions. This reinforces the paper’s central point: organizational form is endogenous.

Table VIII
Probit Estimates for Refocusing Firms

The dependent variable takes the value 1 when the firm operates in multiple segments and 0 otherwise. The reported results are for a sample of single-segment firms and all focusing firms included in the excess value regressions using sales multipliers. Columns 2 to 4 display the estimated coefficients, *t*-statistics, and marginal effects estimated at the mean values of the exogenous variables. The value *EBIT/SALES* is the ratio of *EBIT* to sales, and *CAPX/SALES* is the ratio of capital expenditures to sales. The variables *A_TA*, *A_EBIT*, and *A_CAPX* are the historical average value of log of *TA*, *EBIT/SALES*, and *CAPX/SALES*. The variable *PNDIV* is the fraction of all firms in the industry that are conglomerates, and *PSDIV* is the fraction of industry sales accounted for by conglomerates. The value *MNUM* is the number of announced mergers/acquisitions in the year, while *MVOL* is the U.S. dollar value of these. *GDP (GDP1)* is the growth rate in real GDP (and its lagged value), while *CONTRACTION (CONTRACTION1)* is the number of months in the year the economy was in a recession (and its lagged value). The variable *MAJOREX* is a dummy that takes the value 1 if the firm is listed in Nasdaq, NYSE, or AMEX, and 0 otherwise. The variable *FOREIGN* is a dummy that takes the value 1 if the firm is incorporated outside the United States and 0 otherwise. The variable *SNP* is a dummy variable, which takes the value 1 when the firm is part of the S&P index and 0 otherwise.

	Coefficient	<i>T</i> -statistics	Marginal Effect
Constant	-3.40	-35.21	-0.24
Log of Total Assets	-0.09	-1.53	0.00
<i>EBIT/SALES</i>	-0.28	-1.94	-0.07
<i>CAPX/SALES</i>	0.24	1.58	-0.02
Log of <i>TA</i> (1 lag)	-0.20	-2.29	-0.02
<i>EBIT/SALES</i> (1 lag)	0.05	0.29	-0.03
<i>CAPX/SALES</i> (1 lag)	0.85	4.77	0.01
Log of <i>TA</i> (2 lag)	0.11	1.88	-0.01
<i>EBIT/SALES</i> (2 lag)	-0.14	-1.07	-0.01
<i>CAPX/SALES</i> (2 lag)	0.88	5.20	-0.02
<i>SNP</i>	-0.06	-1.54	0.01
<i>PNDIV</i>	0.02	14.97	0.003
<i>PSDIV</i>	0.01	12.87	0.001
<i>MVOL</i>	0.0003	1.02	0.0002
<i>MNUM</i>	-0.21	-8.10	-0.04
<i>GDP</i>	0.04	4.62	0.01
<i>GDP1</i>	0.01	0.93	-0.001
<i>CONTRACTION</i>	0.32	3.90	0.10
<i>CONTRACTION1</i>	0.25	3.15	0.05
<i>A_TA</i>	0.43	11.56	0.07
<i>A_EBIT</i>	-0.52	-4.40	-0.07
<i>A_CAPX</i>	-3.68	-13.97	-0.11
<i>MAJOREX</i>	0.20	7.94	0.03
<i>FOREIGN</i>	-0.33	-5.64	-0.05
No. of observations			27,995
Maximized likelihood			-8381
Likelihood index			0.21
% of positive observations			12.60

Table VIII: Probit estimates for refocusing firms.

7 Short summary

- The paper asks whether the diversification discount is causal evidence that diversification destroys value.
- It argues that firms choose organizational form, so diversification status is endogenous.
- The data come from Compustat segment files from 1978 to 1996 and contain 8,815 firms and 58,965 firm-years.

- Excess value is the log ratio of actual firm value to imputed stand-alone value based on focused-firm industry multiples.
- Standard OLS regressions reproduce the diversification discount, roughly 11% to 13%.
- Diversifying firms differ systematically from always-focused firms, and they often come from industries with high exit rates.
- Firm fixed effects reduce the estimated discount.
- IV estimates and self-selection corrections often turn the diversification discount into a premium.
- The self-selection coefficient for diversifying firms is negative, indicating that firms more likely to diversify are also more likely to have low value.
- Refocusing firms show the opposite selection pattern: correcting for endogeneity can increase the estimated discount associated with remaining multisegment.
- The results imply that observed discounts around diversification are partly selection effects, not direct evidence of value destruction.

8 Conclusion

8.1 Core conclusion

Campa and Kedia show that the diversification discount is not sufficient evidence that diversification destroys value. Diversified firms are discounted in simple regressions, but firms choose to diversify. Once the endogeneity of that choice is modeled, the estimated discount falls substantially and sometimes becomes a premium.

8.2 Economic intuition

Firms diversify or refocus when their environments change. A firm in a declining industry may diversify because its focused prospects are weak. The market may discount the firm because of those weak prospects, not because diversification itself destroys value. Similarly, a firm may refocus when one segment has strong opportunities and the costs of operating multiple segments become high.

8.3 Incremental contribution revisited

The paper's contribution is to connect the diversification discount to selection. It does not deny that agency costs or inefficient internal capital markets can exist. Instead, it shows that the empirical discount cannot be interpreted causally unless the diversification decision itself is modeled.

8.4 Implications

The findings imply that corporate strategy and firm value should be analyzed jointly. Policies, governance reforms, or managerial conclusions based only on observed discounts can be misleading if they ignore why firms select into diversification or refocusing.

8.5 Limitations

The paper relies on observational data, valid instruments, and selection-model assumptions. Segment data are imperfect, and imputed stand-alone values are only approximations. The paper weakens the simple causal interpretation of the discount, but it does not provide a full dynamic structural model of diversification and refocusing.

8.6 Takeaway

The enduring takeaway is that **firms choose to diversify**. The diversification discount is partly a selection phenomenon: the firms that diversify are different before they diversify, and those differences matter for value. Organizational form should therefore be treated as an endogenous firm choice, not as an exogenous treatment.